

LAB 4 – VIBE CODING

Project: Bubble Bobble

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GitHub Repository: vismayag476-cmd/26_bubbleBobble

https://github.com/vismayag476-cmd/26_bubbleBobble

Chat History – Vibe Coding Tasks

User: I am completing a Vibe Coding lab. Please inspect the existing game.py code and fix ONLY the bubble-trapping bug.

The README says the bug is in trap_enemies: it compares a squared distance against a threshold that was not squared to match it. This causes an enemy to be trapped only when the bubble almost perfectly overlaps it.

Please:

1. Find the exact faulty comparison in the existing code.
2. Fix the squared-distance/threshold mismatch.
3. Make the smallest possible change.
4. Do not change any other game functionality.
5. Explain exactly what you changed and why.

GitHub Copilot: Optimized tool selection! I'll inspect `trap\_enemies` and the README's bug note, then make only the threshold correction and run a narrow syntax check.

Read `[]()`, lines 1 to 260

Read `[]()`

Read memory

The faulty check is `distance\_squared\_to(...) < 24 \* 2`: the left side is squared pixels, while `24 \* 2` is only 48. I'll change only that threshold to `24 \*\* 2` (576), preserving the intended 24-pixel contact radius; a syntax check is the narrow validation.

Generating patch (10 lines) in `[]()`

...

...

Ran `pylanceFileSyntaxErrors`

Completed with input: {

```
"fileUri": "file:///c:/Users/VISMAYA%20G/26_bubbleBobble/game.py",  
"workspaceRoot": "c:\\Users\\VISMAYA G\\26_bubbleBobble"  
}
```

In `game.py`, `trap\_enemies` compared a squared distance with `24 \* 2` (48), instead of the squared 24-pixel contact threshold. I changed only that comparison to `24 \*\* 2` (576), so bubbles can trap enemies within the intended radius.

Validation: `game.py` has no syntax errors.

Made changes.

User: Now implement ONLY the bubble_tint(bubble) feature from the README.

Please inspect the existing game.py first.

Requirements:

- bubble_tint(bubble) is called once per bubble per frame.
- It receives a Bubble object.
- It should return an (r, g, b) tuple or None.
- bubble.enemy tells whether an enemy is trapped.
- bubble.life indicates the bubble's remaining lifetime.
- Make the bubble visually change colour as its remaining life decreases.
- Keep the implementation simple and readable.
- Do not modify unrelated game logic.
- Make the smallest possible change needed.

First explain what you are going to change, then apply the code change.

GitHub Copilot:

User: @agent Try Again

GitHub Copilot: I'll re-read the current `game.py`, especially `bubble\_tint` and the nearby `Bubble` fields, to confirm the latest edits. Then I'll explain the minimal color mapping and change only that function.

Searched for regex `def bubble\_tint|class Bubble|def draw` (`\*\*/game.py`), 3 results

Read , lines 1 to 55

The current function is still a `pass`, and `Bubble.life` is initialized to `BUBBLE\_LIFE` and decreases only while an enemy is trapped. I'll have `bubble\_tint` return `None` for ordinary bubbles and interpolate trapped bubbles from the existing pale-pink tint toward red as `life` runs out; I'll leave the unrelated `zimport random` at the file top untouched.

Read `game.py`, lines 90 to 120

Generating patch (11 lines) in `game.py`

...

...

Ran `pylanceFileSyntaxErrors`

Completed with input: {

 "fileUri": "file:///c:/Users/VISMAYA%20G/26_bubbleBobble/game.py",

 "workspaceRoot": "c:\\Users\\VISMAYA G\\26_bubbleBobble"

}

The tint implementation is in place. The whole-file syntax check is blocked by the existing first-line text `zimport random`, which I've left untouched per the "only this feature" constraint; I'll do a narrow syntax check of the edited function itself.

Ran `pylanceSyntaxErrors`

Completed with input: {

 "code": "def bubble_tint(bubble):\n \"\"\"Return an (r, g, b) colour for a bubble, or None for the default.\"\"\"\n if not bubble.enemy:\n return None\n life_ratio = max(0.0, min(1.0, bubble.life / BUBBLE_LIFE))\n return (255, int(80 + 110 * life_ratio), int(80 + 150 * life_ratio))",

 "pythonVersion": "3.10"

}

Implemented only `bubble\_tint` in `game.py`. Bubbles without a trapped enemy return `None`, preserving their existing default color. Trapped bubbles transition from pale pink to red as `bubble.life` decreases, with the ratio clamped so the RGB values stay in range.

The edited function passes a syntax check. The whole file currently has a pre-existing syntax error at `zimport random` on line 1; I left it unchanged to keep this change limited to the requested feature.

Made changes.

User: import random

GitHub Copilot:

User: @agent Try Again

GitHub Copilot: I'll check the current first line and correct it to the valid ``import random`` statement if the typo is still there.

User: `import random`

GitHub Copilot: The file read didn't return its contents, so I'm checking the opening line again before touching it.

Searched for regex `^(zimport random|import random)$` (**/game.py`)`, 1 result

The file still starts with ``zimport random``, which is invalid Python. I'll fix only that typo to ``import random``.

Generating patch (6 lines) in

...

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Ran ``pylanceFileSyntaxErrors``

Completed with input: {

`"fileUri": "file:///c:/Users/VISMAYA%20G/26_bubbleBobble/game.py",`

`"workspaceRoot": "c:\\Users\\VISMAYA G\\26_bubbleBobble"`

}

Fixed the typo at the top of ``game.py``: ``zimport random`` is now ``import random``. The file passes the syntax check.

Made changes.

User: Now implement ONLY the `on_fruit_collected(fruit)` feature from the README.

Please inspect the existing `game.py` first.

Requirements:

- `on_fruit_collected(fruit)` is called immediately after the player collects a fruit.
- The fruit has already been removed and its points have already been added.

- The function receives the collected Fruit object.
- Its return value is ignored.
- Add a simple visible fruit-collection counter to the HUD so the player can see how many fruits have been collected.
- Keep the implementation simple and consistent with the existing game.
- Do not modify scoring, lives, levels, bubbles, enemies, or unrelated functionality.
- Make the smallest possible change needed.

First explain the change, then apply it.

GitHub Copilot: I'll inspect how fruit is collected, how `Game.reset()` initializes state, and how the HUD is rendered. Then I'll add a counter updated by the callback and display it, keeping gameplay mechanics unchanged.

Searched for regex `def reset|on_fruit_collected|fruit.value|self.font.render` (**/game.py`), 6 results`

Read `game.py`, lines 1 to 60

Fruit collection already calls `on_fruit_collected(fruit)` after adding points and removing the fruit. I'll add a module-level collection count, increment it in that callback, reset it with the game, and append the count to the existing HUD text; scoring and other gameplay logic will remain unchanged.

Generating patch (22 lines) in `game.py`

...

...

Ran `pylanceFileSyntaxErrors`

Completed with input: {

```
  "fileUri": "file:///c:/Users/VISMAYA%20G/26_bubbleBobble/game.py",
  "workspaceRoot": "c:\\Users\\VISMAYA G\\26_bubbleBobble"
}
```

Implemented the fruit counter in `game.py`. The existing callback now increments the count after collection, `Game.reset()` clears it, and the HUD displays `Fruits N`. No scoring or other gameplay behavior was changed. The file passes the syntax check.

Made changes.

User:

Now implement ONLY the `bonus_life_threshold()` feature from the README.

Please inspect the existing `game.py` first.

Requirements:

- The function takes no arguments.
- It should return an integer score threshold or `None`.
- Use a threshold of 5000.
- The existing game bookkeeping already handles awarding the bonus life when the score crosses the threshold.
- Do not modify the existing bonus-life bookkeeping.
- Do not change scoring, lives, levels, bubbles, enemies, or unrelated functionality.
- Make the smallest possible change needed.

Apply the change and verify that the file passes a syntax check.

GitHub Copilot: I'll inspect the current `bonus_life_threshold()` implementation, then change only its return value to `5000` and run a syntax check.

Read `[]()`, lines 15 to 40

`bonus_life_threshold()` currently returns `None` implicitly via `pass`. I'll replace only that `pass` with `return 5000`, leaving the existing award bookkeeping untouched.

Generating patch (8 lines) in `[]()`

...

...

Ran `pylanceFileSyntaxErrors`

Completed with input: {

```
"fileUri": "file:///c:/Users/VISMAYA%20G/26_bubbleBobble/game.py",  
"workspaceRoot": "c:\\Users\\VISMAYA G\\26_bubbleBobble"  
}
```

Updated `bonus_life_threshold()` in `game.py` to return `5000`. No bonus-life bookkeeping or other game logic was changed. The file passes the syntax check.

Made changes.